

SEDAR 25: South Atlantic Black sea bass and golden tilefish

Black sea bass

Data Workshop

Life History

- Investigate the movements and migrations of black sea bass using otolith microchemistry, genetic studies, and expanding tagging studies.
- Investigate the movement and mixing of larval and juvenile black sea bass within the U.S. South Atlantic region.
- Sampling to include the entire Southeast region over a longer time period.
- Analyze size- or age-specific spawning frequency and spawning seasonality.
- Further develop the tagging model described by Rudershausen et al. (2010) to address the assumptions of the model.
- Depth appears to have an effect on the discard mortality rate. Currently depth-specific discard rates and estimates of discard numbers are not available. There is very little depth specific information on the private recreational fleet.
- Temperature and seasonality of discard mortality should be investigated.
- Circle hooks are now required by the SAFMC for fishermen operating in the snapper grouper fishery. The impact of this regulation cannot currently be incorporated into the discard mortality rate.
- Venting is not required in the South Atlantic but it is required in the Gulf of Mexico for snapper grouper fishermen. Research should be conducted on a variety of recompression techniques to determine the most effective method for reducing discard mortality.

Commercial Statistics

- The Commercial Workgroup recommends study of migration patterns, focusing on fish movements around the Cape Hatteras, NC area.
- Additionally, the group would suggest determining the impact/landings of the historical foreign fleet in the South Atlantic.
- Finally, collection of better spatial information in the fishery to determine potential localized depletion effects is recommended.

Recreational Statistics

- Increase sample size of at-sea observers and dockside validation for HB mode.
- Increase proportion of fish with biological data within MRFSS sampling.
- Development of hard part sampling coordinated with intercept surveys.
- Continue development of standardized method for calculating incomplete weight data
- Quantify historical fishing photos for use in future SEDARS.
- Develop method for capturing depth at capture within MRFSS At-Sea observer program and Headboat Survey.
- Conduct study looking at current compliance rates in logbook programs, develop recommendations for improving them, including increased education directed toward effect of not reporting accurately.
- Continued development of electronic reporting of headboat logbook for full implementation

- Continued development of higher degree of information of condition of released fish e.g. FL as the model
- Continued evaluation of methodology for mandatory reporting in the For-hire sector e.g. Gulf MRIP Pilot

Indices

- None submitted.

Assessment Workshop

- The assessment panel recommended increasing the number of age samples collected from the general recreational sector.
- Black sea bass in the southeast U.S. were modeled in this assessment as a unit stock, as recommended by the DW and supported by genetic analysis (SEDAR-25-RD42). For any stock, variation in exploitation and life-history characteristics might be expected at finer geographic scales. Modeling such sub-stock structure would require more data, such as information on the movements and migrations of adults and juveniles, as well as spatial patterns of recruitment. Even when fine-scale spatial structure exists, incorporating it into a model may or may not lead to better assessment results (e.g., greater precision, less bias). Spatial structure in a black sea bass assessment model might range from the very broad (e.g., a single Atlantic stock) to the very narrow (e.g., a connected network of meta-populations living on individual reefs). What is the optimal level of spatial structure to model in an assessment of snapper grouper species such as black sea bass?
- The assessment time period (1978–2010) is short relative to some other assessments of South Atlantic reef fishes. Extending the assessment back in time might provide improved understanding of the stock's potential productivity and therefore sustainable yield, assuming the historic productivity is still relevant. Such an extension would require historic landings estimates from all fleets in operation. Although historic estimates from the commercial sector are available, those from the recreational sector are not. Hindcasting the historic recreational landings might require the development of new methods, or at least analysis of existing methods.
- Protogynous life history: 1) Investigate possible effects of hermaphroditism on the steepness parameter; 2) Investigate the sexual transition for temporal patterns, considering possible mechanistic explanations if any patterns are identified; 3) Investigate methods for incorporating the dynamics of sexual transition in assessment models.
- In this assessment, the number of spawning events per mature female per year assumed a constant value of $X = 31$. That number was computed from the estimated spawning frequency and spawning season duration. If either of those characteristics depends on age or size, X would likely also depend on age or size. For black sea bass, does spawning frequency or spawning season duration (and therefore X) depend on age or size? Such dependence would have implications for estimating spawning potential as it relates to age structure in the stock assessment.

- For this assessment, the age-dependent natural mortality rate was estimated by indirect methods. More direct methods, e.g. tag-recapture, might prove useful. Some tag-recapture studies have demonstrated relatively high tag return rates for black sea bass, at least compared to those of other reef fishes of the southeast U.S.

Review Workshop

The RP was in agreement with the research recommendations from the Data Workshop and Assessment Workshop reports. These identify the main shortcomings in the data and assessment which might be improved by research. It is worth noting that alongside any improvements in methodology and information, allowance should be made for backwards compatibility with existing long time-series. The recommendations are extensive and some priority may be placed so that research having the greatest impact on the assessment might be given the greatest priority.

High Priority

Life history: There are a number of uncertainties over the life history of this species which are critical in setting up reliable age-structured stock assessment models. Any studies that improve understanding of size or age specific spawning frequency, spawning seasonality, and functions modeling sex-change should be given high priority, particularly because they are critical in defining SSB and therefore stock status. This is particularly important in black sea bass because it depends on a calculation of female fecundity where mortality is apparently focused on the males (protogyny with age specific selectivity and low undersize discard mortality).

Ageing: Age data is an important part of the assessment. Where possible, age sampling should be improved in terms of coverage by maximizing the number of trips sampled from both the recreational and commercial landings and discards.

Discards: Discards make up a significant proportion of the catch, but mortality of discards is estimated as low. This mortality estimate is important in the stock assessment, and research to improve its accuracy could have significant impact on the assessment. Studies could improve estimates by relating mortality to temperature and depth and improving the routine collection of temperature and depth data. Also, any improvement on estimates of discards and research that would reduce discard mortality (e.g. hook type, venting) should have high priority.

Recreational Statistics: The RP believed that research recommendations with the objective of improving recreational statistics could have significant impact on the black sea bass stock assessment. Any program to improve recreational fishery data would cover a wide number of other stocks making it efficient. High priority research and data collection should include improvements in the headboat survey, in methods to estimate weight from length, compliance with logbook programs and development of electronic logbooks where appropriate. Also, the improvements would be enhanced with the research on discards, discard mortality and ageing outlined above.

Historical catches: The AW recommended extending the catch history further back than 1978. The RP considers that this is a high priority as it can significantly change the perception of the productivity of the stock. However, it should be noted that any such extension is almost always associated with great uncertainty both in the estimation of historical catch and in the implicit comparison with a historical baseline that might have changed due to climate and other factors.

Medium Priority

Stock structure: A number of research recommendations by the DW and AW indicated possible ways to improve definitions of stock structure (e.g. genetic analyses). The RP found no very significant problem with this issue in this assessment. However, stock structure, including smaller scale spatial structure, movement and resident times could be valuable. The AW also suggested carrying out simulations to look at how spatial data and models might be included in a stock assessment, and the RP agrees that this might be a good start point before more expensive research is undertaken.

Indices: Abundance indices are usually the main information drivers in the stock assessments in these fisheries. The RP recommended improving the fishery independent index if possible, ensuring geographical coverage of the stock is complete. Also, local absolute stock size estimates might be obtained from underwater video surveys, tagging, depletion fishing experiments within a small area, or some combination of these three. Estimating absolute biomass should be done in a way which is informative on catchability and selectivity in the model (could be included as a prior, for example).

Recreational Statistics: Some research on the recreational fishery, while useful, was in the opinion of the RP, less urgent. This included analysis of historical photos to obtain lengths, research to obtain and interpret condition information on discarded fish and the evaluation of some data collection programs.

Life history: The AW recommended looking at estimating age-dependent natural mortality directly. While the RP recognized that natural mortality is an important parameter, estimating this quantity is likely to be very difficult and may not be practical. Similarly, ontogenetic migration and other movement patterns, a possible cause of dome-shaped selectivity and local depletion, could be investigated. If a tagging program was being implemented for other purposes, these issues could and should be included.

Recruitment Patterns: The RP noted that the apparent variance in the recruitment residuals had decreased over time. The recruitments are estimates of the model, so it was not necessarily clear that this was a real change in the stock dynamics, random chance or an artifact of the model. Nevertheless, the RP believed that some simple research to support or discount recruitment change could be undertaken by reviewing recruitment in other stocks or correlating this change with environmental variables where some causal link could be hypothesized.

Low Priority

The Commercial Statistics working group suggested examining the impact of the historical foreign fleet. However, the RP believed that the impact of any activities on black sea bass would be low, obtaining any data would be difficult and could be unsuccessful.

Ultimately the interval between the current and next assessment is a policy decision, requiring scientific input. The RP wants to highlight scientific factors that should be taken into consideration when making this decision. The current black sea bass assessment indicates the stock is not overfished, but not yet rebuilt; and is undergoing overfishing. This indicates the stock is likely in need of regular assessments to track its status, ensure overfishing ends, and the stock is on a trajectory to rebuild. No new data sources are expected to be available, at least in the short term, limiting the utility of conducting a new benchmark assessment in the short term.

If management actions change, conducting a new assessment after their implementation has the potential to identify the impacts of the new management actions on the stock, as well as better identify the stock's dynamics. A new assessment could provide improved information on benchmarks such as MSY or status indicators such as B/B_{MSY} .

The RP recommends that assessment updates be conducted regularly, at the interval of a high risk stock, and more often in response to changes in management regulations. If an update assessment indicates the stock's status is declining or new data become available, the RP recommends moving forward with a full benchmark assessment.

Tilefish

Data Workshop

Life History

- Investigate the movements and migrations of Tilefish using Otolith microchemistry
- Investigate the stock definition through genetic studies to establish if biogeographic boundary exists at Cape Hatteras or if future assessments will use the NC/VA border.
- Fishery-dependent and fishery-independent sampling to include the entire Southeast Region throughout a longer time period.
- Analyze size or age specific spawning frequency and spawning seasonality.

Commercial Statistics

- The Commercial Workgroup recommends exploration of the definition of the stock, particularly with respect to the northern boundary.
- Additionally, the group would suggest examining the impact/landings of the historical foreign fleet in the South Atlantic.
- Finally, collection of better spatial information in the fishery to determine potential localized depletion effects is recommended.

Recreational Statistics

- Continue development of standardized method for calculating incomplete weight data
- Develop method for capturing depth at capture within MRFSS At-Sea observer program and Headboat Survey.
- Conduct study looking at current compliance rates in logbook programs, develop recommendations for improving them, including increased education directed toward effect of not reporting accurately.
- Continued development of electronic reporting of headboat logbook for full implementation
- Continued development of higher degree of information of condition of released fish e.g. FL as the model

Indices

- None provided.

Assessment Workshop

- The assessment panel made the following recommendations.
- Increasing the number of age samples collected from the main part of the species' range
- Investigate reproductive characteristics, particularly regarding whether senescence or hermaphroditism occurs in the species
- Improve the genetic data available by conducting studies of gene similarities by region
- Investigate whether a climate-recruitment link exists
- Investigate whether time varying M may be appropriate for tilefish
- Evaluate patterns in ageing error at the data workshop including development of an ageing error matrix
- Obtain MRIP intercept numbers at the DW for tilefish and other rarely caught species

Review Workshop

The RP was in agreement with the research recommendations from the Data Workshop and Assessment Workshop reports. These identify the main shortcomings in the data and assessment which might be improved by research. However, the recommendations are extensive and some priority may be placed so that research having the greatest impact on the assessment might be given the greatest priority.

High Priority

Life history: There are a number of uncertainties over the life history of this species which are critical in setting up reliable age-structured stock assessment models. Some of this basic information is lacking, such as whether the species exhibits hermaphroditism. For example, in the Gulf of Mexico tilefish assessment (SEDAR 22), protogynous hermaphroditism was included in the model, whereas in this assessment it was not. Any studies that improve understanding of size or age specific spawning frequency, spawning seasonality, and functions modeling sex change should be given high priority, particularly because they are critical in defining SSB and therefore stock status.

Movement: Several recommendations relate to fish movement. The RP recommends research on local population structure related to residence times and local migration, whether by tagging or alternative methods. Understanding fish movement should help understand how catches might cause local depletion and over what area. This could lead to improved data collection and use of spatial data in tractable way within the model.

Indices: Abundance indices are usually the main information drivers in the stock assessments in these fisheries. The RP recommends developing a fishery independent index, which eventually would greatly improve the determination of stock status. Also, local absolute stock size estimates might be obtained from underwater video surveys (e.g. counting fish burrows), tagging, depletion fishing experiments within a small area, or some combination of these three. Estimating absolute biomass should be done in a way which is informative on catchability and selectivity in the model (could be included as a prior, for example). This last method may be particularly suitable for tilefish, which is probably a relatively sedentary species.

Medium Priority

Stock structure: A number of research recommendations by the DW and AW indicate possible ways to improve definitions of stock structure (e.g. genetic analyses). The RP found no very significant problem with this issue in this assessment. However, it may be that tilefish could be included in a wider program looking at stock structure of a variety of species which perhaps could also include Gulf of Mexico as well as the southern North Atlantic.

Recreational Statistics: The RP believed that research recommendations with the objective of improving recreational statistics would most likely have limited impact on the tilefish stock assessment, and hence these only have medium priority. However, any program to improve recreational fishery data is likely to cover a wide number of other stocks where such data may be more critical. Therefore, any such program as a whole may be given high priority.

Low Priority

The Commercial Statistics working group suggested examining the impact of the historical foreign fleet. However, the RP believed that the impact of any activities on tilefish would be low, obtaining data would be difficult and could be unsuccessful.